Exploratory Data Analysis (EDA) Summary Report

1. Introduction

The purpose of this report is to conduct an exploratory data analysis (EDA) on Geldium's delinquency prediction dataset before predictive modeling begins. The analysis focuses on understanding the dataset structure, identifying missing or inconsistent data, detecting anomalies, and uncovering early indicators of delinquency risk. The findings from this report will support Tata iQ's analytics team in refining the delinquency prediction model and improving customer intervention strategies.

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2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights anomalies, duplicates, and inconsistencies observed during the initial review.

Key Dataset Attributes

- **Number of records:** 500 customer records
- **Number of variables:** 19 columns
- **Duplicate records identified:** 0 duplicates detected
- **Data quality status:** Moderate quality with a small number of missing values and minor anomalies

Key Variables

Variable	Description	Data Type
Customer_ID	Unique identifier for each customer	Categorical
Age	Customer age	Numerical
Income	Annual customer income	Numerical
Credit_Score	Customer credit score	Numerical
Credit_Utilization	Percentage of available credit being used	Numerical
Missed_Payments	Number of missed payments	Numerical
Delinquent_Account	Indicator of delinquency status (0/1)	Binary
Loan_Balance	Outstanding loan amount	Numerical
Debt_to_Income_Ratio	Debt relative to income	Numerical
Employment_Status	Employment category	Categorical
Account_Tenure	Number of years/months with the institution	Numerical

Variable	Description	Data Type
Credit_Card_Type	Card category assigned to customer	Categorical
Location	Customer city/location	Categorical
Month_1 – Month_6	Monthly payment status history	Categorical

Initial Observations and Anomalies

- Missing values were identified in Income, Credit_Score, and Loan_Balance.
- Credit utilization values exceeded 100% in a few cases, indicating potential outliers or data entry inconsistencies.
- Delinquency prevalence was relatively low (~16%), suggesting mild class imbalance.
- No duplicate customer records were identified.
- Monthly payment history fields contained mixed payment statuses such as “On-time,” “Late,” and “Missed,” which are likely strong behavioral indicators for delinquency prediction.

Initial Data Quality Summary

The dataset is generally structured and usable for predictive modeling; however, several quality concerns must be addressed before model training. Missing financial variables such as income and loan balance could bias predictions if untreated. Some credit utilization observations exceeded expected thresholds, indicating possible outliers or over-limit credit usage behavior. Despite these issues, the dataset contains valuable behavioral and financial indicators that are suitable for delinquency risk analysis.

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3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the selected treatment methods.

Missing Data Summary

Variable	Missing Values	Treatment Method	Justification
Income	39 missing values	Median Imputation	Median reduces sensitivity to income outliers and preserves distribution stability
Loan_Balance	29 missing values	Median Imputation	Loan balances are skewed, making median more robust than mean
Credit_Score	2 missing values	Mean Imputation	Very low missing rate; mean preserves overall distribution with minimal distortion

Additional Data Quality Actions

- Outlier review was recommended for customers with credit utilization greater than 100%.
- Payment history categories should be standardized and encoded before modeling.
- Continuous variables such as income and loan balance should be normalized or scaled during preprocessing.

Recommended Missing Data Strategy

The preferred approach is to retain most records and use statistical imputation rather than deleting rows, since the percentage of missing data is relatively low. Median imputation is appropriate for skewed financial variables, while mean imputation is sufficient for fields with negligible missingness. Synthetic data generation was considered unnecessary because the missing rates were manageable and traditional imputation methods were adequate.

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4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Relationships between variables and delinquency outcomes were reviewed to uncover insights relevant to predictive modeling.

Key Findings

- Customers with higher credit utilization generally showed slightly higher delinquency rates.
- Higher debt-to-income ratios appeared associated with increased repayment risk.
- Payment history variables (Month_1–Month_6) showed recurring “Late” and “Missed” payment patterns, making them strong behavioral predictors.
- Customers with shorter account tenure may represent slightly higher risk due to limited repayment history.
- Missed payment counts alone did not show a strong linear correlation with delinquency, suggesting interactions with other variables may be important.

Correlation and Pattern Insights

Risk Indicator	Why It Matters
Credit_Utilization	High utilization may indicate financial stress and overreliance on available credit
Debt_to_Income_Ratio	Customers with higher debt burdens may struggle to meet repayment obligations
Payment History (Month_1–Month_6)	Repeated late or missed payments are direct indicators of repayment behavior
Credit_Score	Lower scores generally signal weaker creditworthiness and repayment risk

Risk Indicator	Why It Matters
Account_Tenure	Shorter tenure provides less historical reliability data for assessing repayment behavior

Unexpected Findings and Anomalies

- Correlations between delinquency and several numerical variables were weaker than expected, indicating that non-linear relationships or combined behavioral patterns may play a larger role.
- Some customers with relatively strong income levels still displayed delinquency behavior, suggesting that spending patterns and utilization behavior may be more predictive than income alone.
- Credit utilization values above 1.0 (100%) may indicate over-limit borrowing or data entry issues and should be investigated further.

Early Indicators of Delinquency Risk

- Frequent “Late” and “Missed” payment statuses across multiple months.
- Elevated debt-to-income ratios.
- High credit utilization levels.
- Lower credit scores combined with missed payments.
- Newer accounts with inconsistent payment histories.

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5. AI & GenAI Usage

Generative AI tools were used to accelerate dataset review, summarize patterns, recommend imputation strategies, and identify potential delinquency indicators.

Example AI Prompts Used

- “Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency.”
- “Identify the top 3 variables most likely to predict delinquency based on this dataset and explain why.”
- “Suggest an imputation strategy for missing values in financial datasets based on industry best practices.”
- “Propose best-practice methods to handle missing credit utilization data for predictive modeling.”
- “Highlight potential anomalies that may require investigation before predictive modeling.”

AI-Assisted Insights

- AI-assisted analysis helped quickly identify missing data concentrations and unusual utilization values.
- GenAI recommendations supported the use of median imputation for skewed financial variables.
- AI-generated summaries helped prioritize behavioral indicators such as payment history and credit utilization for downstream modeling.

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6. Conclusion & Next Steps

The EDA revealed that the dataset is generally suitable for predictive modeling but requires targeted preprocessing to improve reliability and accuracy. Key issues included missing values in financial variables, outlier credit utilization values, and potential class imbalance in delinquency outcomes. Behavioral variables such as monthly payment history, credit utilization, and debt-to-income ratio emerged as the most relevant indicators of delinquency risk.

Recommended Next Steps

1. Perform data preprocessing and imputation using the selected methods.
2. Standardize and encode categorical payment history variables.
3. Investigate outlier credit utilization observations.
4. Apply feature engineering to create behavioral risk indicators.
5. Address class imbalance before model training using resampling or weighting techniques.
6. Begin predictive modeling using classification algorithms such as logistic regression, random forest, or gradient boosting.

The cleaned and enriched dataset will provide a stronger foundation for Geldium’s delinquency prediction model and support more effective customer intervention strategies.